

## Fourth day of class - 9/4

**Fun fact:** The number 26 is the only positive integer which is one more than a square and one less than a cube. That is

$$5^2 + 1 = 26 = 3^3 - 1.$$

If  $y^2 + 1 = x^3 - 1$ , then  $y^2 = x^3 - 2$ . This elliptic curve has infinitely many rational points, but the only integer points are  $(3, \pm 5)$ .

**Last time:** We talked about how to determine if a conic has any rational points on it, and when there are rational points, how to parametrize them. We talked some about rational points on curves of genus  $\geq 1$ .

**Plan:** Start talking about adding points on elliptic curves. We'll have to introduce some new concepts (like projective space) to really do it right, but we'll get started.

### Elliptic curves:

Let  $E : y^2 = x^3 + Ax + B$ , where  $A, B \in \mathbb{Z}$ .

If we take the intersection of a line with  $E$ ,  $y = mx + b$ , we get a cubic equation. If we choose the line to go through a rational point  $P$  on  $E$ , even one with rational slope, the line will usually intersect the curve in  $P$ , and in two other points that are both irrational.

Instead, what we do is we take *two* rational points  $P = (c, d)$  and  $Q = (e, f)$  on  $E$  and let  $L$  be the line that goes through both  $P$  and  $Q$ . This line  $y = mx + b$  gives a cubic equation  $(mx + b)^2 = x^3 + Ax + B$ . Two roots of this cubic equation are  $e$  and  $f$ . This means that when we factor out  $(x - e)$  and  $(x - f)$  from the cubic, we're left with a linear factor, which gives a third rational root. Let  $R = (g, h)$  be the third intersection point on the curve. Define  $P + Q = (g, -h)$ . This is called the "group law".

**Example:** Let  $y^2 = x^3 - 16x + 16$ . Let  $P = (1, 1)$  and  $Q = (0, 4)$ . The line through  $P$  and  $Q$  has slope  $(1 - 4)/(1 - 0) = -3$ . It has the equation  $y - 1 = -3(x - 1)$  or  $y = -3x + 4$ . When we plug this in to the equation we get  $(-3x + 4)^2 = x^3 - 16x + 16$ . How do we solve this??

Well, we know that  $x = 0$  and  $x = 1$  are already solutions. When we expand the left hand side we get  $9x^2 + \dots = x^3 - 16x + 16$ , so our cubic equation is  $x^3 - 9x^2 + \dots$ . That's all we need to know!

This factors as  $(x - 0)(x - 1)(x - g)$ , where  $g$  is the  $x$ -coordinate of  $R$ , the third intersection point. When we multiply out this cubic we get  $x^3 - (0 + 1 + g)x^2 + \dots$  and this must equal  $x^3 - 9x^2 + \dots$  and so  $0 + 1 + g = 9$ . Thus,  $g = 8$ . We plug 8 into  $y = -3x + 4$  and get  $(-3)(8) + 4 = -24 + 4 = -20$ . So  $R = (8, -20)$ . Thus,  $P + Q = (8, 20)$ .

Note: We are guaranteed that if  $P$  and  $Q$  are rational points, the third intersection will also have a rational  $x$ -coordinate. Since it is on a line with a rational slope and a rational  $y$ -intercept, the  $y$ -coordinate will be rational as well.

If  $E : y^2 = x^3 + Ax + B$  is an elliptic curve, we let  $E(\mathbb{Q})$  be the set of all rational points on  $E$ . We really *want*  $E(\mathbb{Q})$  to be a group. This means we need the following four properties:

- (a) For all  $P, Q \in E(\mathbb{Q})$ , then  $P + Q \in E(\mathbb{Q})$ .
- (b) For all  $P, Q, R \in E(\mathbb{Q})$ , we have  $(P + Q) + R = P + (Q + R)$ .
- (c) There is a special point  $\mathcal{O}$  so that for all  $P \in E(\mathbb{Q})$ ,  $P + \mathcal{O} = P$ . (In order for this to happen, we need to work in projective space.)
- (d) For all  $P \in E(\mathbb{Q})$ , there is a point  $-P \in E(\mathbb{Q})$  so that  $P + (-P) = \mathcal{O}$ . In our case, if  $P = (x, y)$ , then  $-P = (x, -y)$ , but even defining  $P + (-P)$  requires thinking a bit about projective space.

### Another viewpoint

Let  $C$  be any cubic curve, defined by  $f(x, y) = 0$  (so cubic means that  $f(x, y)$  involves at least one term of degree 3), and let  $C(\mathbb{Q})$  be the set of rational points on  $C$ .

Suppose that  $\mathcal{O}$  is a rational point on  $C$ . If  $P$  and  $Q$  are rational points on  $C$ , then we let  $P * Q$  be the third intersection point of the line through  $P$  and  $Q$  with the curve  $C$ . This will again be a rational point.

Now, we define  $P + Q = \mathcal{O} * (P * Q)$ . (Draw a picture.) I claim now that all four properties we want above are satisfied.

Property 1: We know that if  $P, Q \in C(\mathbb{Q})$ , then  $P * Q \in C(\mathbb{Q})$ . Since  $\mathcal{O} \in C(\mathbb{Q})$ , then  $P + Q = \mathcal{O} * (P * Q)$  is in  $C(\mathbb{Q})$  too.

Property 3: By definition, we have  $P + \mathcal{O} = \mathcal{O} * (P * \mathcal{O})$ . There are three points on the line through  $P$  and  $\mathcal{O}$ , namely  $P$ ,  $\mathcal{O}$  and  $P * \mathcal{O}$ . Thus, the third intersection of the line through  $\mathcal{O}$  and  $P * \mathcal{O}$  is  $P$ , and so  $\mathcal{O} * (P * \mathcal{O}) = P$ . Thus,  $P + \mathcal{O} = P$ .

Property 4: Draw the tangent line to the curve  $C$  and  $\mathcal{O}$ . This line intersects the curve  $C$  “twice” at  $\mathcal{O}$  and so this line intersects the curve again at a point  $S \neq \mathcal{O}$ . Given  $P$ , we define  $-P = P * S$ . Now, the line through  $P$  and  $S$  intersects the curve three times: at  $P$ , at  $S$  and at  $P * S$ . Thus,

$$P + (-P) = \mathcal{O} * (P * (P * S)).$$

We have that  $P * (P * S) = S$ . So  $P + (-P) = \mathcal{O} * S$ . Now,  $\mathcal{O}$  has multiplicity two, so the third intersection of the line through  $\mathcal{O}$  and  $S$  is just  $\mathcal{O}$ . Hence  $P + (-P) = \mathcal{O}$ .

### Associativity

We need to show that  $(P + Q) + R = P + (Q + R)$ . To do this, we need to use Bézout’s theorem.

Bézout’s theorem states that if  $C_1$  is a curve of degree  $m$  and  $C_2$  is a curve of degree  $n$ , then  $C_1$  and  $C_2$  intersect in exactly  $mn$  points (counting multiplicity, and these intersection points might occur over  $\mathbb{C}$ ).

(Historical note: This theorem appeared explicitly in Newton’s Principia in 1687. The first proof was given by Étienne Bézout in 1779, but the proof was in the generic case where all multiplicities are one. Properly stating the theorem in all cases and proving it requires an

algebraic definition of multiplicities which only came in 1958 from Jean-Pierre Serre. See Section A.4 of the text for this definition.)

Application: Suppose that  $C$ ,  $C_1$  and  $C_2$  are three cubic curves. By Bézout's theorem,  $C_1$  and  $C_2$  intersect in nine points. Suppose that  $C$  goes through eight of the nine intersection points of  $C_1$  and  $C_2$ . Then  $C$  goes through the ninth one as well.

Proof (sort of): A cubic curve is defined by a polynomial with 10 coefficients. If we multiply all the coefficients by a constant, we get the same curve, so there's really a "nine-dimensional space" of cubic curves. Each time we require the curve to go through a point, that imposes a linear relation on the coefficients. So there is a 1-dimensional space of cubic curves  $C$  that goes through the first eight intersection points. If  $F_1$  and  $F_2$  are the cubic polynomials that define  $C_1$  and  $C_2$ , then we also get a one-dimensional family of curves  $\lambda_1 F_1 + \lambda_2 F_2 = 0$ . I claim these are the same, and so the curve  $C$  has an equation of the form  $\lambda_1 F_1 + \lambda_2 F_2$ . This implies that all nine intersection points of  $F_1$  and  $F_2$  are on  $C$ . QED

### Proof of associativity:

To prove that  $(P + Q) + R = P + (Q + R)$ , we need  $\mathcal{O} * ((P + Q) * R) = \mathcal{O} * (P * (Q + R))$ , so it suffices to prove that  $(P + Q) * R = P * (Q + R)$ .

Key idea: We can apply Bézout's theorem for cubic curves that are degenerate. If we take the equations of three lines and multiply them together, we get a cubic curve!

Draw a picture - Figure 1.9 (page 14) from the text.

**Curve  $C_1$ :** Let  $C_1$  be the union of three lines. The first is the line through  $Q$  and  $R$ . The second is the line through  $O$  and  $P * Q$ . The third is the line through  $P$  and  $Q + R$ .

The first line has  $Q$ ,  $R$  and  $Q * R$  on it.

The second line has  $O$ ,  $P * Q$  and  $P + Q$  on it.

The third line has  $P$ ,  $Q + R$  and  $P * (Q + R)$  on it.

**Curve  $C_2$ :** Let  $C_2$  be the union of three lines. The first is the line through  $P$  and  $Q$ . The second is the line through  $O$  and  $Q * R$ . The third is the line through  $R$  and  $P + Q$ .

The first line has  $P$ ,  $Q$ , and  $P * Q$  on it.

The second line has  $O$ ,  $Q * R$  and  $Q + R$  on it.

The third line has  $R$ ,  $P + Q$  and  $R * (P + Q)$  on it.

These two cubic curves overlap in  $O$ ,  $P$ ,  $Q$ ,  $R$ ,  $P * Q$ ,  $Q * R$ ,  $Q + R$ ,  $P + Q$ . Thus, both curves must go through the same ninth point. This point is  $P * (Q + R) = R * (P + Q)$ . Associativity!

### What's next

When we have a curve of the form  $y^2 = x^3 + ax + b$ , what's  $\mathcal{O}$ ?